

F200 / FLUID FILTER

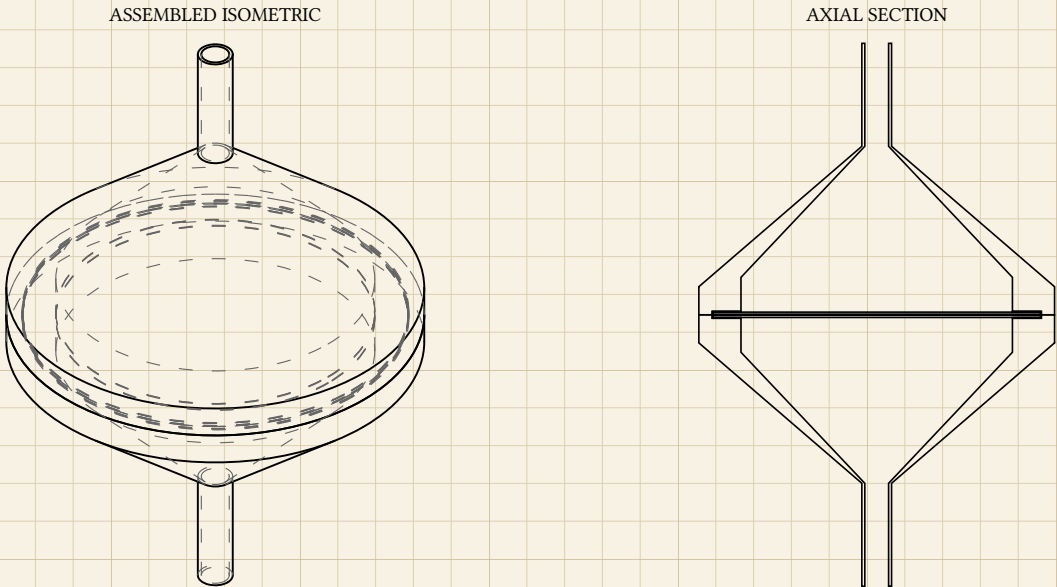
Seven components. One pressure-boundary weld.

PROJECT	Fluid filtration
DOCUMENT	F200-COMP-001
AUTHOR	Engineering concept
CHECKED	Review required
REV	A
DATE	2026-09-21

1 Design intent

Two identical CP titanium Grade 2 cups close on a five-layer stainless pack: frame, coarse support, Dutch-twill cloth, coarse support, frame. One orbital weld is the pressure-boundary closure. The inlet and outlet are integral stubs that share an outside diameter and a wall; the bore is that diameter minus twice the wall. Fluid properties, filtration rating and the sweep points are workbook inputs. Finished-pack resistance is not calibrated. The reference-cloth curves and the loading curves are sensitivities. They do not qualify clean loss, loaded loss, retention or burst.

ASSEMBLED FILTER AND AXIAL SECTION



2 Requirements and interpretation

Targets below are the F100 values carried into this component. The inlet is specified as outside diameter and wall rather than as a bore. A local stress screen or a CAD mass is not a leak, proof, burst or retention test. Those requirements stay open until the planned tests record evidence.

REQUIREMENTS

ID	Requirement	Target
F-01	Operate at MEOP without leakage.	650 psi
F-02	Withstand proof pressure without leakage or unacceptable permanent deformation.	1300 psi
F-03	Withstand the specified minimum burst pressure.	2600 psi
F-04	Meet the selected filtration rating and particle retention efficiency.	5 μm ; 99.9 % at 3 μm
F-05	Meet clean pressure-drop limit at nominal flow; characterize surge flow.	5 psi at 0.14 lb/s; surge 0.47 lb/s
F-06	Retain specified dirt mass within loaded pressure-drop limit at nominal flow.	0.318 g; 18 psi at 0.14 lb/s
F-07	Meet dry assembly mass limit.	1 lb
F-08	Approve compatibility and cleanliness for the selected fluid and operating state.	Selected fluid and state
F-09	Demonstrate media and support integrity at blocked-screen differential.	OPEN
F-10	Use the seven-component architecture.	7 parts

REFERENCES

- [1] Convair Division, Low Gravity Propellant Control Using Capillary Devices in Large Scale Cryogenic Vehicles: Related IRAD Studies, General Dynamics / NASA, 1970, GDC-DDB70-009, Eq. 2-2, Table 2-2, Sec. 2.1.7 — Reference Dutch-twill coefficients. Not a calibration for the selected filtration rating. ntrs.nasa.gov/api/citations/19710001290/download...
- [2] ATI, ATI CP Grade 2 titanium, ATI, Minimum tensile properties — Yield 275 MPa; ultimate 345 MPa. Confirm procurement form, temperature and weld allowables. www.atimaterials.com/Products/Pages/ati-cp2.aspx
- [3] IAPWS, IAPWS formulation for water viscosity, IAPWS, 2008, Default fluid example — Ambient-pressure water properties carried over from F100. Properties at operating pressure require confirmation. iapws.org/relguide/visc.pdf
- [4] Project engineering, Filter design inputs and pressure-drop method, Project record, 2026, F100 baseline and supplied September 2026 method — Pressure, flow and material concept inherited. Inlet OD and wall changed by user. Geometry and mesh packaging remain preliminary.

3 Calculation constants

Selected fluid: Water (reference example) (liquid). Ambient-pressure properties at 70 F inherited from F100; operating-pressure correction OPEN. Use density and viscosity at the selected temperature and pressure. This is a single-phase, constant-property model; significant gas density change, two-phase flow and non-Newtonian behavior need a different flow model. Fluid / material compatibility remains OPEN. Edit `input/design.xlsx`, then run `uv_run_doc.py`. The constants drive the CAD and every calculation below. Change density, viscosity and temperature together when the fluid state changes. ^[3]

CONSTANTS

Symbol	Value / unit	Definition / basis
T	70 °F	Selected fluid temperature; User
ρ	997.99 kg/m ³	Selected fluid density; Default water example at 70 F, ambient pressure. Replace for selected fluid and operating state.
μ	0.000979 Pa·s	Selected fluid dynamic viscosity; Default water example at 70 F, ambient pressure. Replace for selected fluid and operating state.
$\dot{m}_n$	0.14 lb/s	Nominal mass flow; User
$\dot{m}_s$	0.47 lb/s	Surge mass flow; User
p_{meop}	650 psi	Housing operating pressure difference; User: internal minus external
p_{proof}	1300 psi	Housing proof pressure difference; User
p_{burst}	2600 psi	Housing burst pressure difference; User
$\Delta p_{\text{p, clean}}$	5 psi	Clean pressure-drop limit; User
$\Delta p_{\text{p, loaded}}$	18 psi	Loaded pressure-drop limit; User; nominal flow basis
m_d	0.318 g	Retained dry dirt mass; User; contaminant unspecified

Symbol	Value / unit	Definition / basis
m_{limit}	1 lb	Maximum dry hardware mass; User
x_{rating}	5 μm	Filtration rating target; User; test definition pending
η	99.9 %	Capture efficiency target; User
x_{eff}	3 μm	Efficiency particle size; User
d_{face}	58 mm	Frame opening / exposed screen diameter; Design
d_{pack}	70 mm	Frame and mesh outside diameter; Design
t_{frame}	0.2 mm	Uncoined sheet thickness; Design
$t_{\text{assembled}}$	0.17 mm	Nominal coined frame envelope; Unvalidated manufacturing assumption
t_{coarse}	0.48 mm	Each coarse envelope thickness; Calender target; raw catalog 0.508 mm
t_{fine}	0.18 mm	Dutch-twill envelope thickness; Packaging / mass placeholder. Supplier confirmation OPEN for selected finished cloth.
d_{body}	76 mm	Housing outside diameter; Simplified design
d_{pocket}	70.4 mm	Single frame seat outer diameter; Design
z_{outer}	6 mm	Outer cone start from weld plane; Design
z_{inner}	8 mm	Inner cone start from weld plane; Design
z_{stub}	36 mm	Cone end / straight stub start; Design
L_{half}	58 mm	Housing half length; Design
d_{stub}	0.25 in	Inlet and outlet OD; User design input; ID = OD - 2 wall
ρ_{Ti}	4510 kg/m ³	CP titanium Grade 2 density; TIMET
ρ_{SS}	8000 kg/m ³	Stainless frame density; Engineering nominal
S_y	275 MPa	Ti Grade 2 minimum yield screening value; TIMET / ATI; stock certificate required
S_u	345 MPa	Ti Grade 2 minimum tensile screening value; ATI
w_{coarse}	1.367 kg/m ²	Each coarse cloth areal mass; TWP 0.280 lb/ft ² rounded
w_{fine}	0.95 kg/m ²	Dutch-twill areal mass; Packaging / mass placeholder. Supplier confirmation OPEN for selected finished cloth.
f_{growth}	1.2	Dry mass development reserve multiplier; Design assumption
t_{stub}	0.025 in	Inlet and outlet wall thickness; User design input; ID is derived

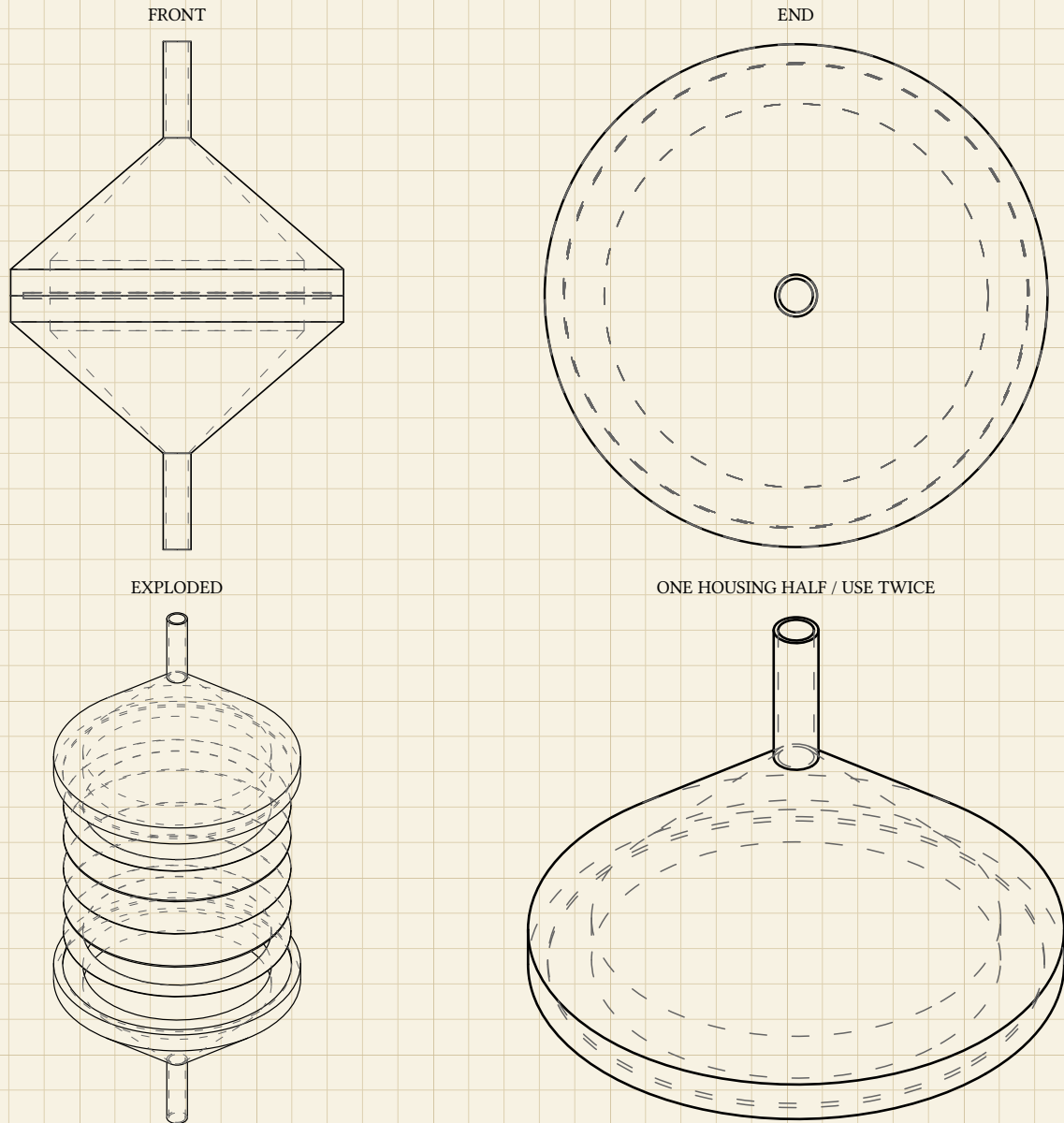
4 Architecture

Seven solids: two identical turned cups, two annular frames, two coarse supports and one fine cloth. There is no thermal moat and no separate seal ring. The pack is trapped on the flat land. In the model the media are envelope solids, so coining, edge sealing and the blocked-screen support capacity remain open.

SEVEN COMPONENTS

Component	Qty	Material / definition
Turned cup	2	CP Ti Grade 2; identical halves with integral stubs
Annular frame	2	316L; raw 0.2 mm, assembled 0.17 mm
Coarse support	2	316 stainless, 40 x 40; assembled 0.48 mm
Fine cloth	1	316 stainless Dutch twill; 5 μm ; thickness 0.18 mm provisional

CAD DRAWING SHEET

**4.1 Interfaces**

Exposed face 58 mm; pack OD 70 mm; body OD 76 mm. Stub OD 0.25 in and wall 0.025 in. Mass uses the revolved housing section, the raw frame volume and the provisional cloth areal weights.

5 Flow basis

Exposed area is the frame opening. Approach velocity follows continuity over that area, $U = \frac{\dot{m}}{\rho A_{\text{face}}}$. The stub bore is derived; it is not a second input. ^[4]

WATER AND GEOMETRY

Symbol	Value / unit	Definition / basis
ρ	997.99 kg/m ³	Selected fluid density; Default water example at 70 F, ambient pressure. Replace for selected fluid and operating state.
μ	0.000979 Pa·s	Selected fluid dynamic viscosity; Default water example at 70 F, ambient pressure. Replace for selected fluid and operating state.
T	70 °F	Selected fluid temperature; User
d_{face}	58 mm	Frame opening / exposed screen diameter; Design
d_{stub}	0.25 in	Inlet and outlet OD; User design input; ID = OD - 2 wall
t_{stub}	0.025 in	Inlet and outlet wall thickness; User design input; ID is derived
$\dot{m}_n$	0.14 lb/s	Nominal mass flow; User
$\dot{m}_s$	0.47 lb/s	Surge mass flow; User

EXPOSED AREA AND APPROACH VELOCITY

$$d_i = d_o - 2 \cdot t = 0.250 \text{ in} - 2 \cdot 0.025 \text{ in} = 0.200 \text{ in} \quad (1)$$

$$A_{\text{face}} = \pi \cdot \frac{(d_{\text{face}})^2}{4} = \pi \cdot \frac{(58.000 \text{ mm})^2}{4} = 2642.079 \text{ mm}^2 \quad (2)$$

$$U_n = \frac{\dot{m}_n}{\rho \cdot A_{\text{face}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.024 \frac{\text{m}}{\text{s}} \quad (3)$$

$$U_s = \frac{\dot{m}_s}{\rho \cdot A_{\text{face}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.081 \frac{\text{m}}{\text{s}} \quad (4)$$

$$U_s = 0.081 \text{ m/s}$$

STUB BORE VELOCITY AND DYNAMIC PRESSURE

$$A_{\text{bore}} = \pi \cdot \frac{(d_i)^2}{4} = \pi \cdot \frac{(0.200 \text{ in})^2}{4} = 20.268 \text{ mm}^2 \quad (5)$$

$$V_n = \frac{\dot{m}_n}{\rho \cdot A_{\text{bore}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 20.268 \text{ mm}^2} = 3.139 \frac{\text{m}}{\text{s}} \quad (6)$$

$$V_s = \frac{\dot{m}_s}{\rho \cdot A_{\text{bore}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 20.268 \text{ mm}^2} = 10.540 \frac{\text{m}}{\text{s}} \quad (7)$$

$$q_n = \rho \cdot \frac{(V_n)^2}{2} = 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(3.139 \frac{\text{m}}{\text{s}})^2}{2} = 0.713 \text{ psi} \quad (8)$$

$$q_s = \rho \cdot \frac{(V_s)^2}{2} = 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(10.540 \frac{\text{m}}{\text{s}})^2}{2} = 8.039 \text{ psi} \quad (9)$$

$$q_s = 8.039 \text{ psi}$$

6 Pressure drop

Clean loss of one screen is the viscous-plus-inertial form $\Delta p_0 = C_v \mu U + C_i \rho U^2$, with $C_v = A_1 \frac{L}{D_a^2}$ and $C_i = A_2 \frac{L}{D_a}$. D_a is the measured average capillary diameter, not the filtration rating. No extra porosity correction is applied. ^[1]

The three published rows are individual clean cloths. A finished pack has to include both supports and the way the stack is held. That calibration is open on every rating. The worked calculation is the tightest published cloth, 200 x 1400. The table repeats nominal and surge loss for all three.

SCREEN RESISTANCE

$$dp_{\text{screen}} = C_i \cdot \rho \cdot U^2 + C_v \cdot U \cdot \mu \quad (10)$$

PUBLISHED REFERENCE CLOTHS

Cloth	Da (µm)	L (mm)	A1	A2
200 x 1400	22.800	0.14732	190	18
165 x 1400	28.300	0.15240	150	16
200 x 600	36.700	0.13970	52	3

200 X 1400 CLEAN SCREEN

$$C_v = A1 \cdot \frac{L}{(Da)^2} = 190.000 \cdot \frac{0.147 \text{ mm}}{(22.800 \text{ µm})^2} = 53845029.240 \frac{1}{\text{m}} \quad (11)$$

$$C_i = A2 \cdot \frac{L}{Da} = 18.000 \cdot \frac{0.147 \text{ mm}}{22.800 \text{ µm}} = 116.305 \quad (12)$$

$$dp_n = C_v \cdot \mu \cdot U_n + C_i \cdot \rho \cdot (U_n)^2 \quad (13)$$

$$= 53845029.240 \frac{1}{\text{m}} \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.024 \frac{\text{m}}{\text{s}} + 116.305 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \left(0.024 \frac{\text{m}}{\text{s}}\right)^2 \quad (14)$$

$$= 0.194 \text{ psi} \quad (15)$$

$$dp_s = C_v \cdot \mu \cdot U_s + C_i \cdot \rho \cdot (U_s)^2 \quad (16)$$

$$= 53845029.240 \frac{1}{\text{m}} \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.081 \frac{\text{m}}{\text{s}} + 116.305 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \left(0.081 \frac{\text{m}}{\text{s}}\right)^2 \quad (17)$$

$$= 0.728 \text{ psi} \quad (18)$$

$$dp_s = 0.728 \text{ psi}$$

REFERENCE-CLOTH LOSS ON THE EXPOSED FACE

Cloth	Nominal loss	Surge loss
200 x 1400	0.194 psi	0.728 psi
165 x 1400	0.105 psi	0.409 psi
200 x 600	0.019 psi	0.073 psi

6.1 Housing loss

Housing loss is separate from the cloth: $\Delta p_{\text{housing}} = K \rho \frac{V_{\text{box}}^2}{2}$. K is not taken from a fitting table or a test. The worked line uses $K = 1$. The table lists every workbook value, and the inlet plots use the largest of those values and say so.

HOUSING LOSS AT $K = 1$

$$dp_{\text{housing}_n} = K \cdot q_n = 1.000 \cdot 0.713 \text{ psi} = 0.713 \text{ psi} \quad (19)$$

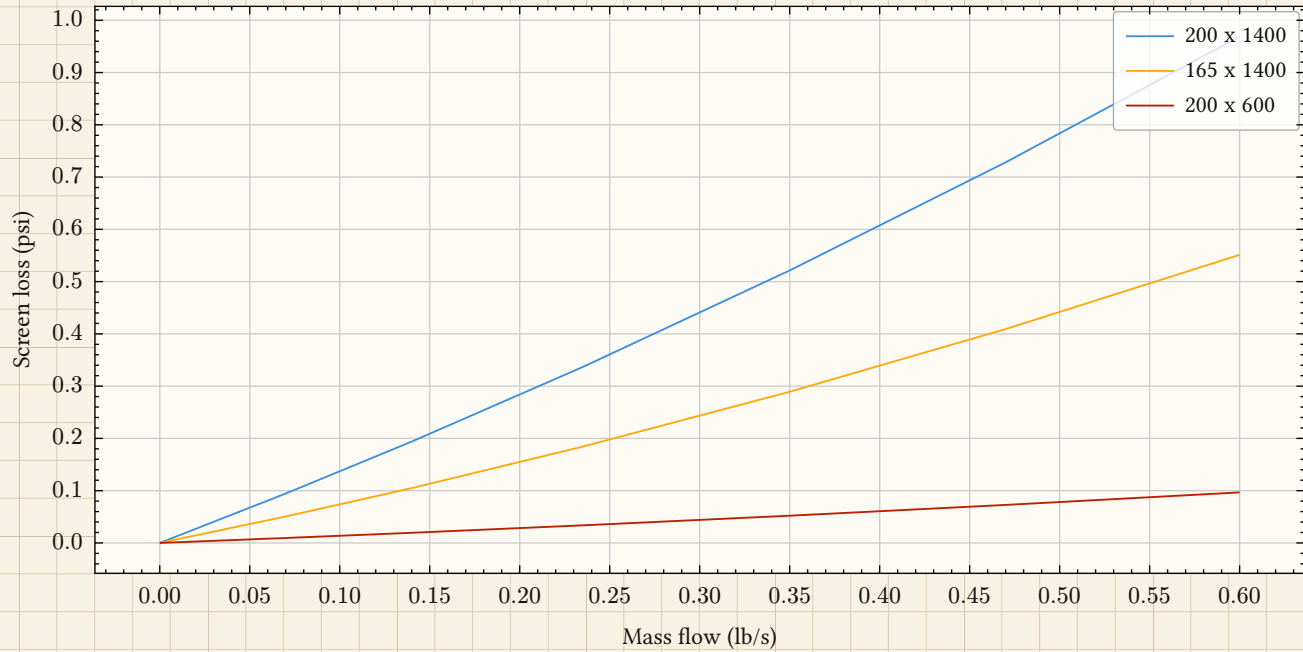
$$dp_{\text{housing}_s} = K \cdot q_s = 1.000 \cdot 8.039 \text{ psi} = 8.039 \text{ psi} \quad (20)$$

$$dp_{\text{housing}_s} = 8.039 \text{ psi}$$

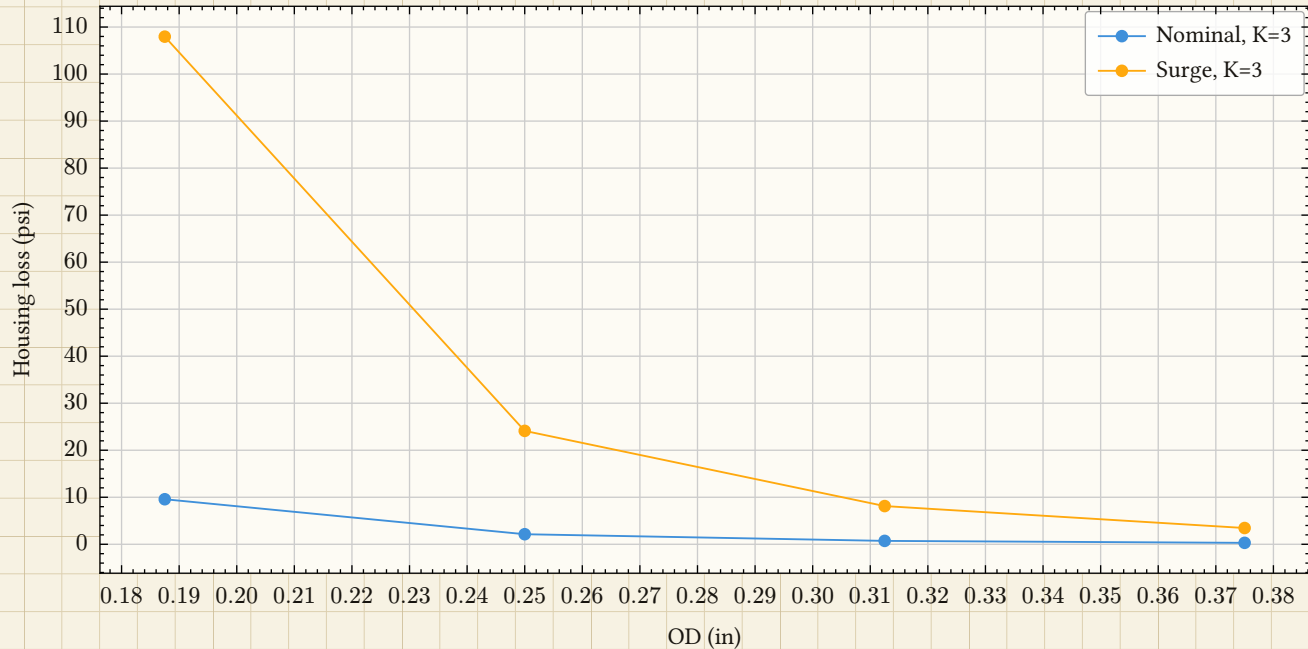
HOUSING LOSS SENSITIVITY

Assumed K	Nominal loss	Surge loss
0	0 psi	0 psi
1	0.713 psi	8.039 psi
3	2.14 psi	24.118 psi

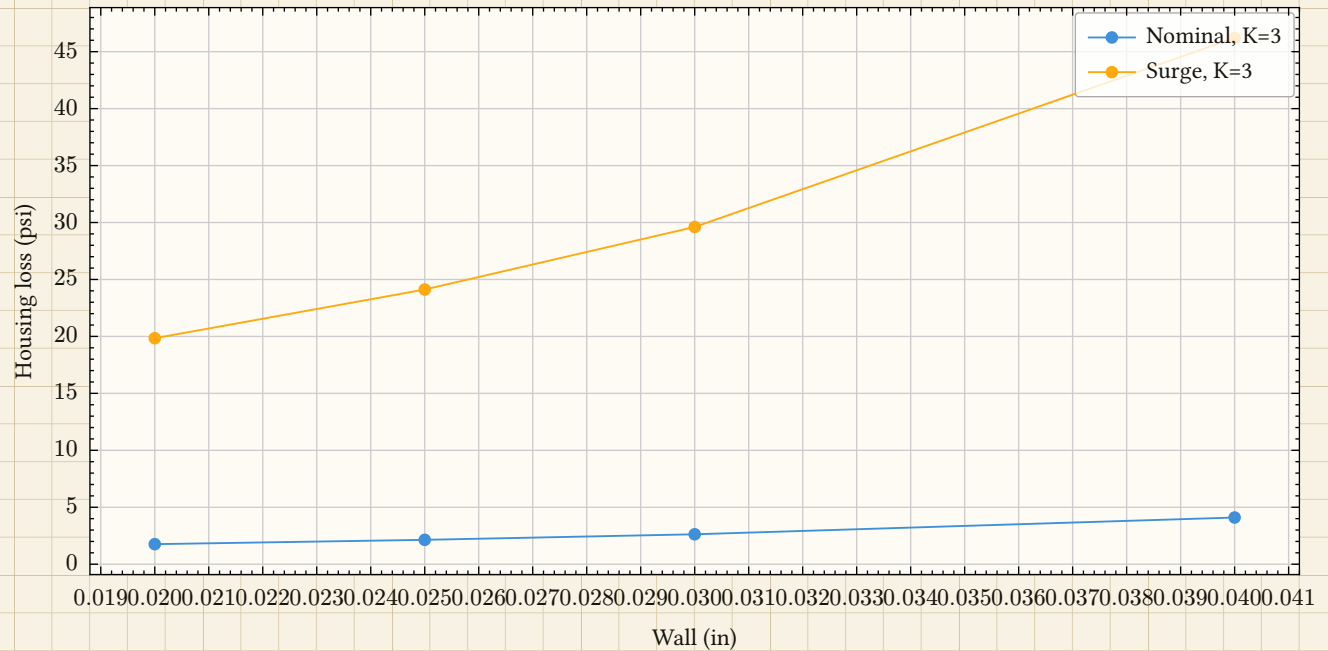
FLOW SENSITIVITY — REFERENCE SCREENS ONLY



INLET OD SENSITIVITY — ASSUMED K



INLET WALL SENSITIVITY – ASSUMED K



6.2 Loading

Blockage and cake are separate idealizations. For a blocked-area fraction b , $F = 1 - b$ and $\Delta p_b = C_v \mu \frac{U}{F} + C_i \rho \frac{U^2}{F^2}$. The worked blockage is half the face. A cake instead adds $\mu U \alpha_c \frac{m_d}{A_{\text{face}}}$ to the clean loss of the same cloth. The worked cake resistance is the 10^9 m/kg sweep point. Blockage is not inferred from dirt mass. Neither curve establishes dirt capacity, retention or media strength. ^[4]

The clean and loaded limits apply at nominal flow. Surge is shown because the workbook asks for it. No separate surge loss limit is specified.

LOSS LIMITS AND RETAINED DIRT

Symbol	Value / unit	Definition / basis
Δp_{clean}	5 psi	Clean pressure-drop limit; User
Δp_{loaded}	18 psi	Loaded pressure-drop limit; User; nominal flow basis
m_d	0.318 g	Retained dry dirt mass; User; contaminant unspecified
x_{rating}	5 μm	Filtration rating target; User; test definition pending
η	99.9 %	Capture efficiency target; User
x_{eff}	3 μm	Efficiency particle size; User

HALF-FACE BLOCKAGE – 200 X 1400

$$F = 1 - b = 1 - 0.500 = 0.500 \quad (21)$$

$$\Delta p_{\text{blocked}} = C_v \cdot \mu \cdot \frac{U_n}{F} + C_i \cdot \rho \cdot \frac{(U_n)^2}{(F)^2} \quad (22)$$

$$= 53845029.240 \frac{1}{\text{m}} \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot \frac{0.024 \frac{\text{m}}{\text{s}}}{0.500} + 116.305 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(0.024 \frac{\text{m}}{\text{s}})^2}{(0.500)^2} \quad (23)$$

$$= 0.407 \text{ psi} \quad (24)$$

$$\Delta p_{\text{blocked}} = 0.407 \text{ psi}$$

CAKE AT 1E9 M/KG – 200 X 1400

$$dp_{cake} = \mu \cdot U_n \cdot \alpha \cdot \frac{m_d}{A_{face}} = 0.001 \text{ Pa} \cdot \text{s} \cdot 0.024 \frac{\text{m}}{\text{s}} \cdot 1000000000.000 \frac{\text{m}}{\text{kg}} \cdot \frac{0.318 \text{ g}}{2642.079 \text{ mm}^2} = 0.412 \text{ psi} \quad (25)$$

$$dp_{loaded} = dp_{clean} + dp_{cake} = 0.194 \text{ psi} + 0.412 \text{ psi} = 0.605 \text{ psi} \quad (26)$$

$$dp_{loaded} = 0.605 \text{ psi}$$

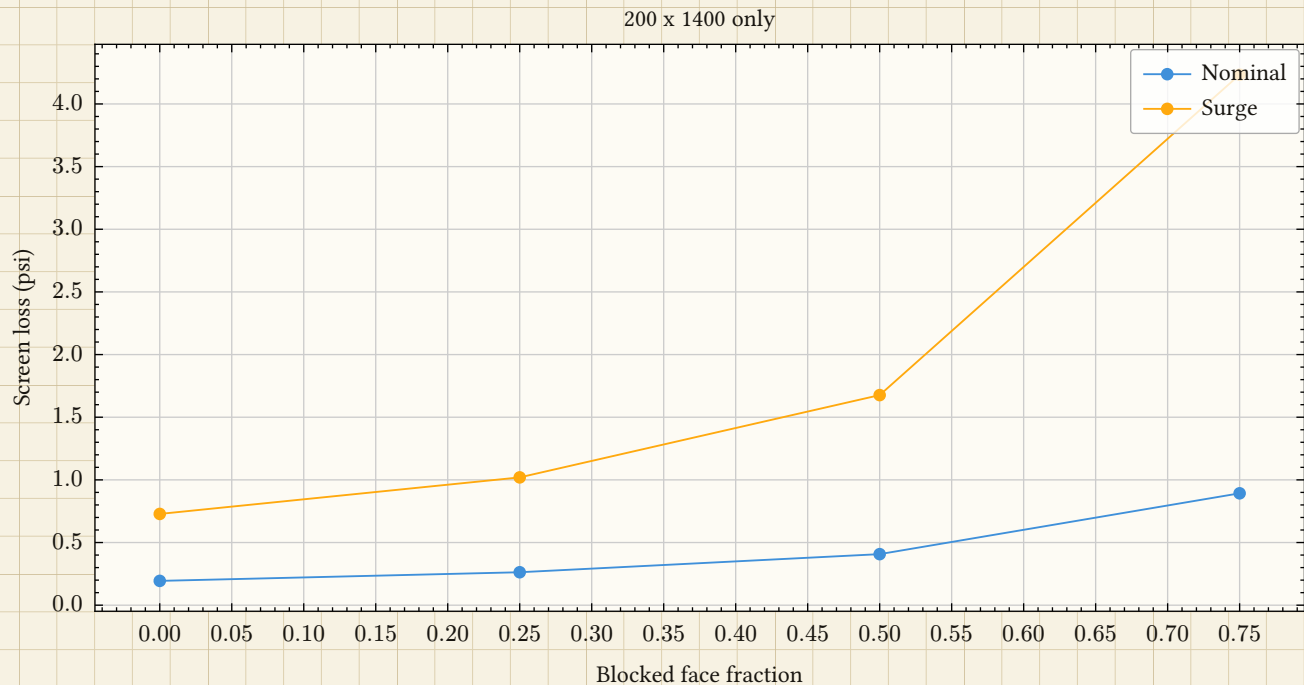
6.3 What these sensitivities say

The 200 x 1400 cloth, alone, loses 0.194 psi at nominal flow and 0.728 psi at surge. The other two published cloths are lower. All three sit under the nominal clean limit of 5 psi. They are single screens. They are not the finished pack.

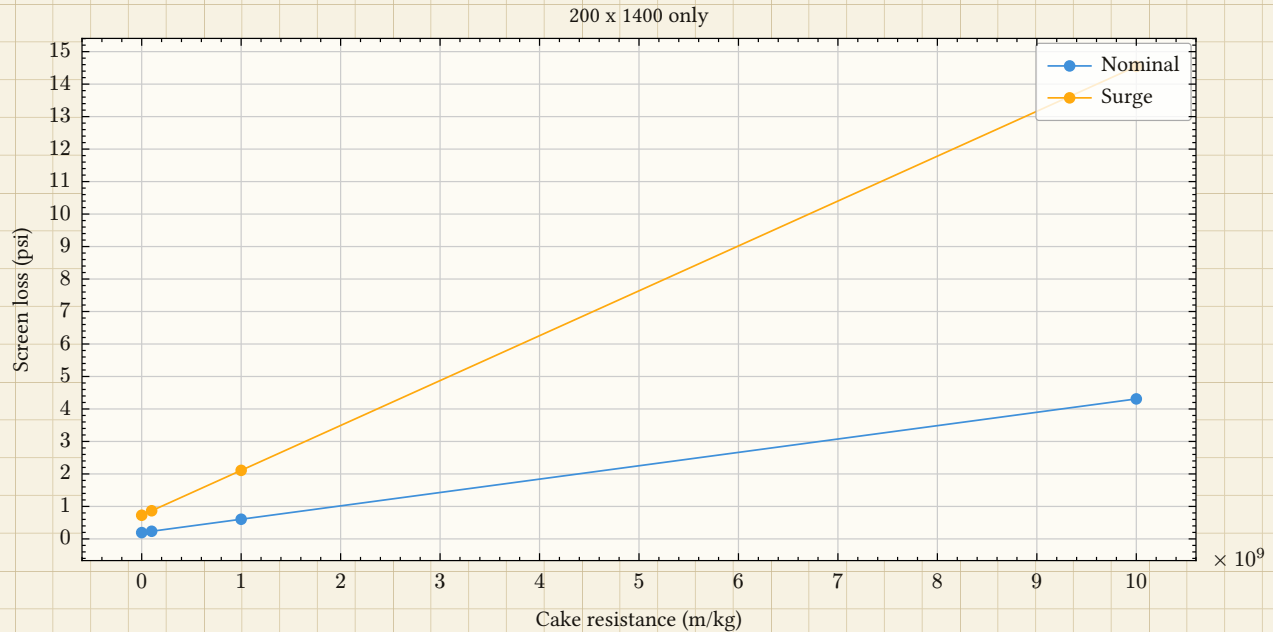
Housing loss at $K = 1$ is the dynamic pressure: 0.713 psi nominal and 8.04 psi at surge. Surge has no separate limit. At $K = 1$ the surge housing loss is already above the nominal clean limit, before any cloth is added.

Half the 200 x 1400 face blocked raises that cloth's nominal loss to 0.407 psi. The illustrative cake at 10^9 m/kg adds 0.412 psi and brings the same cloth to 0.605 psi, under the loaded limit of 18 psi. Both numbers move when the cloth, the supports or α_c change.

BLOCKED-AREA SENSITIVITY – REFERENCE CLOTH



CAKE SENSITIVITY — ASSUMED RESISTANCE



6.4 Calibration

Fit measured clean-pack pressure drop to the two terms μU and ρU^2 at a known fluid state. Record C_v , C_i and the evidence for each rating. Micron rating alone does not supply a resistance curve. Fit housing loss on its own, or measure the complete assembly and say which scope that test covers.

FINISHED-PACK CALIBRATION

Rating	Nominal pack loss	Calibration
1 μm	OPEN	Cv / Ci and evidence missing
3 μm	OPEN	Cv / Ci and evidence missing
5 μm	OPEN	Cv / Ci and evidence missing
10 μm	OPEN	Cv / Ci and evidence missing
20 μm	OPEN	Cv / Ci and evidence missing

7 Housing and mass

The local screen is the closed-end thick cylinder at the bore, $\sigma_{VM} = \frac{\sqrt{3}pr_o^2}{r_o^2 - r_i^2}$. Proof stress is compared with the minimum yield strength and burst stress with the minimum tensile strength. The stub uses the derived bore. The weld rim uses the body outside diameter and the pocket diameter. The cone, the transitions, the weld and fatigue are outside this comparison. It does not qualify proof or burst. Material minima: ^[2].

PRESSURE AND WALL SECTION

Symbol	Value / unit	Definition / basis
p_{meop}	650 psi	Housing operating pressure difference; User: internal minus external
p_{proof}	1300 psi	Housing proof pressure difference; User
p_{burst}	2600 psi	Housing burst pressure difference; User
d_{stub}	0.25 in	Inlet and outlet OD; User design input; ID = OD - 2 wall
t_{stub}	0.025 in	Inlet and outlet wall thickness; User design input; ID is derived
d_{body}	76 mm	Housing outside diameter; Simplified design
d_{pocket}	70.4 mm	Single frame seat outer diameter; Design
S_y	275 MPa	Ti Grade 2 minimum yield screening value; TIMET / ATI; stock certificate required
S_u	345 MPa	Ti Grade 2 minimum tensile screening value; ATI

CLOSED-END CYLINDER SCREEN

$$\sigma_{stub_{proof}} = \sqrt{3} \cdot p_{proof} \cdot \frac{(d_o)^2}{(d_o)^2 - (d_i)^2} \quad (27)$$

$$= \sqrt{3} \cdot 1300.00 \text{ psi} \cdot \frac{(0.25 \text{ in})^2}{(0.25 \text{ in})^2 - (0.20 \text{ in})^2} \quad (28)$$

$$= 43.12 \text{ MPa} \quad (29)$$

$$\sigma_{stub_{burst}} = \sqrt{3} \cdot p_{burst} \cdot \frac{(d_o)^2}{(d_o)^2 - (d_i)^2} \quad (30)$$

$$= \sqrt{3} \cdot 2600.00 \text{ psi} \cdot \frac{(0.25 \text{ in})^2}{(0.25 \text{ in})^2 - (0.20 \text{ in})^2} \quad (31)$$

$$= 86.25 \text{ MPa} \quad (32)$$

$$\sigma_{rim_{proof}} = \sqrt{3} \cdot p_{proof} \cdot \frac{(d_{body})^2}{(d_{body})^2 - (d_{pocket})^2} \quad (33)$$

$$= \sqrt{3} \cdot 1300.00 \text{ psi} \cdot \frac{(76.00 \text{ mm})^2}{(76.00 \text{ mm})^2 - (70.40 \text{ mm})^2} \quad (34)$$

$$= 109.38 \text{ MPa} \quad (35)$$

$$\sigma_{rim_{burst}} = \sqrt{3} \cdot p_{burst} \cdot \frac{(d_{body})^2}{(d_{body})^2 - (d_{pocket})^2} \quad (36)$$

$$= \sqrt{3} \cdot 2600.00 \text{ psi} \cdot \frac{(76.00 \text{ mm})^2}{(76.00 \text{ mm})^2 - (70.40 \text{ mm})^2} \quad (37)$$

$$= 218.75 \text{ MPa} \quad (38)$$

$$MS_{stub_{proof}} = \frac{S_y}{\sigma_{stub_{proof}}} - 1 = \frac{275.00 \text{ MPa}}{43.12 \text{ MPa}} - 1 = 5.38 \quad (39)$$

$$MS_{stub_{burst}} = \frac{S_u}{\sigma_{stub_{burst}}} - 1 = \frac{345.00 \text{ MPa}}{86.25 \text{ MPa}} - 1 = 3.00 \quad (40)$$

$$MS_{rim_{proof}} = \frac{S_y}{\sigma_{rim_{proof}}} - 1 = \frac{275.00 \text{ MPa}}{109.38 \text{ MPa}} - 1 = 1.51 \quad (41)$$

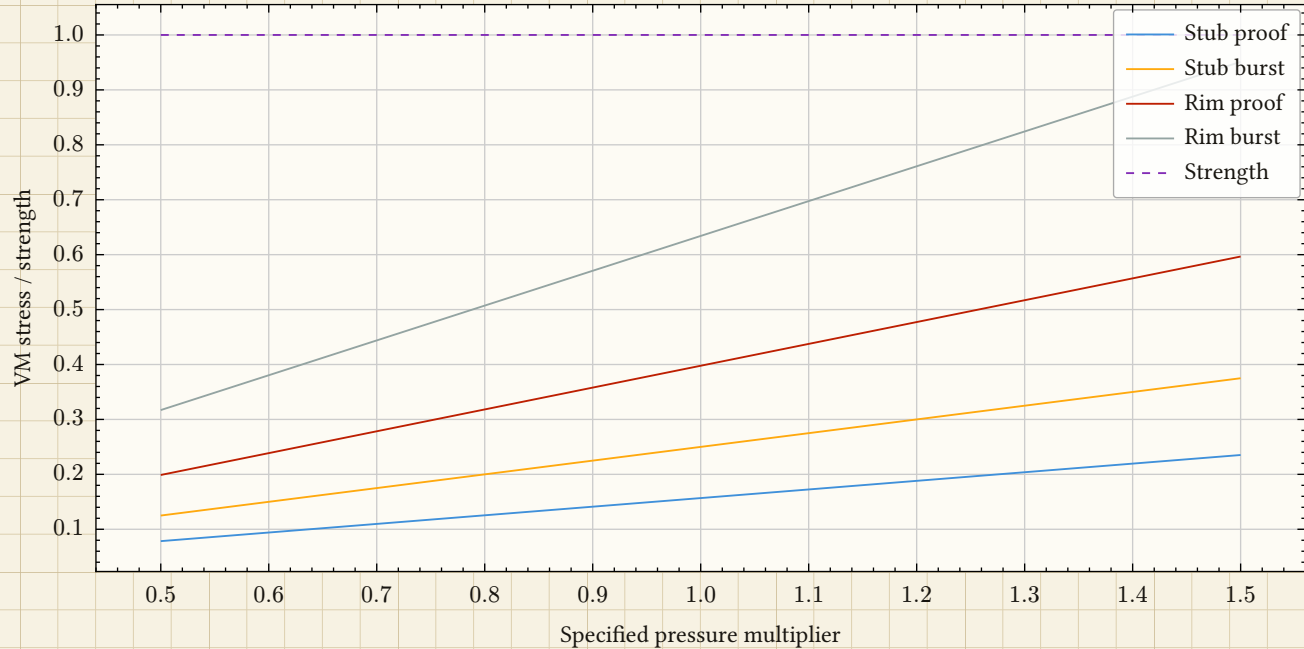
$$MS_{rim_{burst}} = \frac{S_u}{\sigma_{rim_{burst}}} - 1 = \frac{345.00 \text{ MPa}}{218.75 \text{ MPa}} - 1 = 0.58 \quad (42)$$

$$MS_{rim_{burst}} = 0.58$$

7.1 Local margins

Every margin above is positive, so this elastic screen is inside the minimum strengths. The stub is comfortable: proof margin 5.38, burst margin 3.00. The weld rim is the closer section: proof margin 1.51, burst margin 0.58. The cone, the weld and fatigue are not in these numbers.

PROOF AND BURST PRESSURE SENSITIVITY



7.2 Dry mass

Housing volume is the revolved half-section, checked against the CAD solid. Frame mass uses the raw sheet, not the coined envelope. Fine-cloth thickness and areal mass are still placeholders. The growth factor is a development reserve, not a measured tolerance.

MASS INPUTS

Symbol	Value / unit	Definition / basis
ρ_{Ti}	4510 kg/m ³	CP titanium Grade 2 density; TIMET
ρ_{SS}	8000 kg/m ³	Stainless frame density; Engineering nominal
d_{pack}	70 mm	Frame and mesh outside diameter; Design
t_{frame}	0.2 mm	Uncoined sheet thickness; Design
w_{coarse}	1.367 kg/m ²	Each coarse cloth areal mass; TWP 0.280 lb/ft ² rounded
w_{fine}	0.95 kg/m ²	Dutch-twill areal mass; Packaging / mass placeholder. Supplier confirmation OPEN for selected finished cloth.
f_{growth}	1.2	Dry mass development reserve multiplier; Design assumption
m_{limit}	1 lb	Maximum dry hardware mass; User

DRY MASS WITH THE DEVELOPMENT RESERVE

$$A_{pack} = \pi \cdot \frac{(d_{pack})^2}{4} = \pi \cdot \frac{(70.000 \text{ mm})^2}{4} = 3848.451 \text{ mm}^2 \quad (43)$$

$$V_{frame} = \pi \cdot \left((d_{pack})^2 - (d_{face})^2 \right) \cdot \frac{t_{frame}}{4} = \pi \cdot ((70.000 \text{ mm})^2 - (58.000 \text{ mm})^2) \cdot \frac{0.200 \text{ mm}}{4} = 241.274 \text{ mm}^3 \quad (44)$$

$$m_{housings} = 2 \cdot V_{half} \cdot \rho_{ti} = 2 \cdot 27870.808 \text{ mm}^3 \cdot 4510.000 \frac{\text{kg}}{\text{m}^3} = 251.395 \text{ g} \quad (45)$$

$$m_{frames} = 2 \cdot V_{frame} \cdot \rho_{ss} = 2 \cdot 241.274 \text{ mm}^3 \cdot 8000.000 \frac{\text{kg}}{\text{m}^3} = 3.860 \text{ g} \quad (46)$$

$$m_{coarse} = 2 \cdot A_{pack} \cdot w_{coarse} = 2 \cdot 3848.451 \text{ mm}^2 \cdot 1.367 \frac{\text{kg}}{\text{m}^2} = 10.522 \text{ g} \quad (47)$$

$$m_{fine} = A_{pack} \cdot w_{fine} = 3848.451 \text{ mm}^2 \cdot 0.950 \frac{\text{kg}}{\text{m}^2} = 3.656 \text{ g} \quad (48)$$

$$m_{dry} = m_{housings} + m_{frames} + m_{coarse} + m_{fine} = 251.395 \text{ g} + 3.860 \text{ g} + 10.522 \text{ g} + 3.656 \text{ g} = 0.594 \text{ lb} \quad (49)$$

$$m_{reserve} = \text{growth} \cdot m_{dry} = 1.200 \cdot 0.594 \text{ lb} = 0.713 \text{ lb} \quad (50)$$

$$\text{MS}_{mass} = \frac{m_{limit}}{m_{reserve}} - 1 = \frac{1.000 \text{ lb}}{0.713 \text{ lb}} - 1 = 0.403 \quad (51)$$

$$\text{MS}_{mass} = 0.403$$

7.3 Mass margin

The two cups are most of the mass. Dry mass is 0.594 lb; with the 1.2 reserve it is 0.713 lb, against a limit of 1 lb (margin 0.40). The fine-cloth areal weight is still a placeholder, so this is an estimate, not a weighed assembly.

8 Manufacturing

The process sheet is the planned sequence. An acceptance sentence here is not a completed inspection record.

PROCESS

id	operation	acceptance
M1	Turn two identical CP Ti Grade 2 cups with integral stubs	Verify OD, wall, bore, lands and weld preparation against released drawing
M2	Cut 316L annular frames and 316 stainless support / Dutch-twill cloth	Confirm alloy, thickness, rating, areal mass and finished-pack flow curve
M3	Assemble frame / support / fine / support / frame stack	Inspect edge capture and bypass seal. Coining and pack compression require development
M4	Orbital weld the titanium closure seam	Qualify procedure, shielding, inspection and leak acceptance
M5	Clean and identify the assembly	Confirm cleanliness and material compatibility for the selected fluid

9 Planned verification

Empty result and evidence cells stay open. Entering a procedure in the workbook does not close the requirement it names.

PLANNED TESTS

id	requirement	procedure	criterion
T1	F-01	Leak test at MEOP	No leakage at specified MEOP
T2	F-02	Proof pressure test	No leakage or unacceptable permanent deformation at proof pressure
T3	F-03	Destructive burst test	Withstand minimum specified burst pressure
T4	F-04	Particle challenge / retention test	Confirm rating and efficiency at specified particle size
T5	F-05	Clean flow bench at controlled fluid state	Meet nominal clean loss limit; characterize through surge flow
T6	F-06	Dirt loading at nominal flow	Retain specified dirt mass within loaded loss limit
T7	F-07	Weigh final dry assembly	Meet dry mass limit
T8	F-08	Fluid compatibility and cleanliness review	Approve selected fluid, state and all wetted materials
T9	F-09	Blocked-screen differential test	Define required differential; verify support and media integrity

10 Compliance

OPEN 9 requirement(s) need verification.

COMPLIANCE

ID	Requirement	Assessment	Status
F-01	Operate at MEOP without leakage.	Awaiting test evidence.	OPEN
F-02	Withstand proof pressure without leakage or unacceptable permanent deformation.	Awaiting test evidence.	OPEN
F-03	Withstand the specified minimum burst pressure.	Awaiting test evidence.	OPEN
F-04	Meet the selected filtration rating and particle retention efficiency.	Awaiting test evidence.	OPEN
F-05	Meet clean pressure-drop limit at nominal flow; characterize surge flow.	Finished-pack and housing calibration OPEN.	OPEN
F-06	Retain specified dirt mass within loaded pressure-drop limit at nominal flow.	Loading and retention test evidence OPEN.	OPEN
F-07	Meet dry assembly mass limit.	CAD estimate available; finished mass and cloth areal weights OPEN.	OPEN
F-08	Approve compatibility and cleanliness for the selected fluid and operating state.	Awaiting inspection evidence.	OPEN
F-09	Demonstrate media and support integrity at blocked-screen differential.	Required differential and support capacity OPEN.	OPEN
F-10	Use the seven-component architecture.	7 == 7; Generated seven-solid CAD assembly	PASS